AIDCF

AI Data Centre Facilities Assurance Framework

THREAT, HAZARD & FAILURE MODEL

Version 1.0

Capability-first adverse-condition, failure-propagation and resilience analysis

Artifact ID: AIDCF-THF-001
Normative foundation standard
Draft for controlled review
Author: Siva Sethumadhavan
Release 1: Foundational Methodology

Identify credible adversity. Separate cause from failure. Trace propagation. Validate safeguards safely. Preserve UNKNOWN.

Document control

Field	Value
Title	AIDCF Threat, Hazard & Failure Model v1.0
Artifact ID	AIDCF-THF-001
Status	Draft for controlled review
Depends on	AIDCF Manifesto v1.0; Core Conceptual Model v1.0; Infrastructure Ontology & Taxonomy v1.0
Purpose	Define threats, hazards, causes, failure modes, propagation, consequence, safeguards, residual exposure and recovery.
Users	Owners, operators, engineers, assessors, AI infrastructure teams, security, resilience, risk and assurance practitioners.
Safety boundary	Does not replace licensed engineering or authorise destructive, hazardous or operationally disruptive testing.

Executive thesis

AIDCF determines whether credible adverse conditions can defeat required infrastructure functions, propagate through dependencies and fault domains, overcome safeguards, and prevent a Capability Instance from achieving its Service Outcomes. Component failure is not the endpoint. Capability consequence, residual state, recovery and evidence are.

Canonical chain

Context → Threat / Hazard / Degradation → Initiating Event → Failure Cause → Failure Mode → Local Effect → Dependency Effect → Propagated Failure → Capability Consequence → Safeguard Response → Residual State → Recovery → Assurance Decision

Contents

1. Authority, scope and conformance

2. Core ontology and distinctions

3. Scenario and propagation architecture

4. Threat taxonomy

5. Hazard taxonomy

6. Failure-cause taxonomy

7. Failure-mode taxonomy

8. Effects and consequences

9. Safeguards and resilience

10. Analysis techniques

11. Criticality and uncertainty

12. Eight-domain failure catalogue

13. Cross-domain scenario library

14. Assessment records and evidence

15. Machine-readable model

16. Governance and acceptance

17. Appendices

Part I - Authority, scope and conformance

1. Purpose

This standard provides a controlled, vendor-neutral method for identifying, representing, analysing and recording adverse conditions affecting AI data-centre capabilities across the eight AIDCF domains.

2. Normative requirements

THF-REQ-001 Every material scenario shall identify a Capability Instance and Service Outcome.

THF-REQ-002 Threat, hazard, initiating event, cause, failure mode, effect and consequence shall remain distinct.

THF-REQ-003 Each scenario shall declare boundary, architecture baseline, operating state and operating envelope.

THF-REQ-004 Propagation shall use explicit dependencies and fault domains from AIDCF-003.

THF-REQ-005 Safeguard existence shall not imply effectiveness without applicable evidence.

THF-REQ-006 UNKNOWN shall remain visible for any material unverified path element.

THF-REQ-007 Critical safety analysis shall be performed or approved by competent authorised professionals.

THF-REQ-008 Testing shall be proportionate, authorised and safe.

THF-REQ-009 Excluded and external dependencies shall remain visible.

THF-REQ-010 Average scores shall not override critical gates or catastrophic consequences.

3. Conformance levels

Level	Name	Expectation
F0	Vocabulary aligned	Uses canonical meanings.
F1	Scenario structured	Records event, mode, effects and consequence.
F2	Dependency traced	Maps path, fault domains and states.
F3	Evidence gated	Links safeguards, tests, confidence and UNKNOWN.
F4	Machine actionable	Uses controlled identifiers and validation shapes.

Part II - Core ontology and distinctions

Term	Canonical definition
Threat	Deliberate actor, capability, intent or circumstance capable of adverse infrastructure consequence.
Hazard	Non-deliberate source, condition or event with potential for injury, damage, disruption or environmental impact.
Degradation mechanism	Process progressively reducing condition, margin or performance.
Initiating event	Event or condition starting a specific scenario.
Failure cause	Circumstance or mechanism producing or contributing to a failure mode.
Failure mode	Manner in which a required function is lost, degraded, delayed, incorrect, intermittent or uncontrolled.
Local effect	Immediate result at the failed item or function.
Next-level effect	Effect on containing subsystem or dependent function.
End effect	Effect on Capability Instance or Service Outcome.
Propagation path	Ordered sequence through which effects move across entities and dependencies.
Consequence	Realised or foreseeable effect on capability and stakeholders.
Predisposing condition	Condition increasing susceptibility without necessarily initiating failure.
Safeguard	Measure intended to anticipate, prevent, detect, contain, respond, recover or adapt.
Barrier	Safeguard positioned on a defined path.
Residual condition	State remaining after expected safeguards act.

Term	Canonical definition
Recovery path	Sequence restoring an approved service and infrastructure state.
Latent failure	Failure not detected until another event, demand or test.
Common-cause failure	Multiple failures arising from one shared cause.
Common-mode failure	Nominal alternatives fail similarly or through a shared dependency.
Near miss	Event capable of material consequence that did not realise it.

4. Mandatory distinctions

Do not collapse	Reason
Threat / hazard	Deliberate and non-deliberate sources require different assumptions.
Cause / failure mode	Why it failed differs from how the function failed.
Mode / effect	Failure behaviour differs from resulting condition.
Effect / consequence	Technical effect differs from capability impact.
Fault / failure	A fault can exist without defeating function.
Event / state	Occurrence differs from continuing condition.
Exposure / likelihood	Contact differs from probability.
Detection / prevention	Observing does not stop failure.
Failover / recovery	Transfer differs from restoration.
Redundancy / resilience	Duplication does not prove sustained outcome.
UNKNOWN / low likelihood	Missing evidence is not evidence of improbability.

Part III - Scenario and propagation architecture

5. Bidirectional analysis

Approach	Question	Output
Bottom-up	How can this function fail?	Mode, cause and local/global effects.
Top-down	What can defeat the outcome?	Top event and causal combinations.
Path-based	How can effects traverse dependencies?	Ordered path and breakpoints.
State-based	What changes during maintenance or recovery?	State-specific exposure.
Capability-based	What minimum outcome must remain?	Consequence and graceful-degradation boundary.

6. Path types

Path type	Meaning
Local	Contained within intended fault domain.
Cascading	One failure triggers additional failures.
Common-cause	One cause affects multiple paths.
Common-mode	Similar alternatives fail similarly.
Sequential	Initial change enables later failure.
Compound	Multiple events or states combine.
Latent	Hidden condition activated on demand.
Maintenance-induced	Work or bypass creates exposure.
Recovery-induced	Restoration action creates or reintroduces failure.
Cross-domain	Path traverses multiple AIDCF domains.
Cross-boundary	Path crosses organisation or assessment scope.
Slow-burn	Degradation accumulates to threshold.
Intermittent	Repeated instability complicates diagnosis.

7. Scenario minimum record

Field	Requirement
scenarioId	Required, or explicit UNKNOWN where permitted.
capabilityInstance	Required, or explicit UNKNOWN where permitted.
serviceOutcomes	Required, or explicit UNKNOWN where permitted.
architectureBaseline	Required, or explicit UNKNOWN where permitted.
assessmentBoundary	Required, or explicit UNKNOWN where permitted.
initialOperatingState	Required, or explicit UNKNOWN where permitted.
threatOrHazard	Required, or explicit UNKNOWN where permitted.
initiatingEvent	Required, or explicit UNKNOWN where permitted.
predisposingConditions	Required, or explicit UNKNOWN where permitted.
failureCauses	Required, or explicit UNKNOWN where permitted.
initialFailureMode	Required, or explicit UNKNOWN where permitted.
orderedPathSteps	Required, or explicit UNKNOWN where permitted.
faultDomains	Required, or explicit UNKNOWN where permitted.
safeguards	Required, or explicit UNKNOWN where permitted.
safeguardDependencies	Required, or explicit UNKNOWN where permitted.
residualState	Required, or explicit UNKNOWN where permitted.
capabilityConsequence	Required, or explicit UNKNOWN where permitted.
recoveryPath	Required, or explicit UNKNOWN where permitted.
evidence	Required, or explicit UNKNOWN where permitted.
confidenceAndUnknown	Required, or explicit UNKNOWN where permitted.
criticalGate	Required, or explicit UNKNOWN where permitted.
owners	Required, or explicit UNKNOWN where permitted.
reassessmentTrigger	Required, or explicit UNKNOWN where permitted.

Part IV - Threat taxonomy

Source family	Examples
External actor	Trespasser, saboteur, criminal, activist, competitor or state-aligned actor.
Insider	Employee, contractor, supplier staff or coerced individual.
Supply chain	Manufacturer, integrator, logistics, maintenance, software or firmware supplier.
Tenant/customer	Authorised party acting maliciously, negligently or outside agreement.
Remote operator	Authorised or compromised remote support.

Source family	Examples
Automated source	Malicious code or automation creating infrastructure effect.
Unattributed	Evidence supports deliberate activity but source is unresolved.

Threat event	Infrastructure interpretation
Unauthorised entry	Access to site, plant, hall, cage or rack.
Physical interference	Tampering, switching, obstruction or disconnection.
Theft/removal	Removal of equipment, media, cable, fuel or spares.
Sabotage	Deliberate disabling or damage.
Utility manipulation	Interruption or corruption of power, water, fuel or telecom.
OT compromise	Manipulation of BMS, EPMS, SCADA, controllers or management.
Configuration manipulation	Unauthorised change to settings, thresholds, routes or images.
Supply-chain compromise	Counterfeit, modified, malicious or unsupported component.
Resource exhaustion	Intentional consumption of compute, power, cooling, storage or network.
Evidence destruction	Alteration of logs, telemetry, video or records.
Multi-vector attack	Coordinated physical, cyber, supplier and social actions.

Part V - Hazard taxonomy

Hazard family	Examples
Meteorological	Extreme heat, humidity, wind, lightning, rain, drought or haze.
Hydrological	Flood, storm surge, groundwater, drainage failure or water scarcity.
Geological	Earthquake, settlement, subsidence or soil instability.
Environmental	Particulate, corrosion, pollution, salt or biological contamination.
External fire/industrial	Adjacent fire, smoke, explosion, vibration or vehicle impact.
Electrical	Arc, short circuit, overcurrent, overvoltage, backfeed or stored energy.
Thermal	Overtemperature, heat accumulation, transient or thermal runaway.
Pressure/fluid	Overpressure, rupture, cavitation, leak or contamination.
Structural/mechanical	Overload, vibration, anchorage failure, obstruction or collapse.
Chemical/material	Coolant, refrigerant, battery, fuel, suppressants or incompatibility.
Human factors	Fatigue, workload, interface ambiguity or lost situational awareness.
Control interaction	Bad sensor data, unsafe logic, setpoint conflict or unstable automation.
Utility/geopolitical	Grid, water, fuel, telecom, border, sanctions or civil disruption.

Part VI - Failure-cause taxonomy

Cause family	Examples
Design	Requirement, topology, sizing, protection, tolerance or integration.
Manufacturing	Defect, material variation, contamination or counterfeit.
Installation	Assembly, routing, termination, torque, cleanliness or labelling.
Commissioning	Incomplete scope, unrealistic load or unresolved defect.
Configuration	Setpoint, firmware, driver, route, threshold or compatibility.
Operation	Incorrect/delayed action, bypass, overload or deviation.
Maintenance	Missed/incorrect work, calibration, spares or temporary state.
Ageing	Wear, corrosion, fatigue, insulation, battery degradation or drift.
Capacity	Demand exceeds usable/resilient capacity or reserve is committed.
Environment	Temperature, humidity, fluid quality, particulate, vibration or flood.
Dependency	Upstream, shared, hidden, recovery-only or external reliance.
Control/monitoring	Sensor, logic, alarm, communication, historian or response failure.
Human/organisational	Competence, staffing, fatigue, authority or knowledge concentration.
Supplier/contract	Service, SLA, lead time, access or support failure.
Security	Unauthorised access, manipulation, sabotage or theft.
Change/transition	Expansion, migration, upgrade, rollback or temporary state.
Recovery	Wrong sequence, corrupt baseline, insufficient spare or failed restart.
UNKNOWN	Cause cannot be established with available evidence.

THF-REQ-018 Do not terminate analysis at human error without examining systemic contributors.

THF-REQ-019 Stop decomposition only where further analysis would not change treatment or accountability.

THF-REQ-020 Record alternate hypotheses and excluded causes.

THF-REQ-021 Preserve the observed facts when cause is UNKNOWN.

Part VII - Failure-mode taxonomy

Failure mode	Meaning
Complete loss	Function unavailable
Failure to start	Does not initiate on demand
Failure to stop	Continues when cessation required
Failure to transfer	Alternate path not established
Failure to isolate	Fault crosses boundary
Failure to reset/recover	Cannot return to approved state
Degraded output	Below required capacity or quality
Excess output	Unsafe output exceeds limit
Incorrect output	Wrong value, route, state or target
Delayed output	Too late for outcome
Premature output	Before required conditions
Intermittent operation	Repeated appearance/disappearance
Unstable/oscillating	Uncontrolled variation
Capacity exhaustion	Demand exceeds capacity
Performance collapse	Nonlinear degradation after threshold
Data/configuration corruption	State becomes inconsistent
Loss of visibility	Cannot observe/diagnose
False indication	Indicates absent condition
Failure to alarm	Warning absent or late
Nuisance alarm	False/excessive alarms mask events
Latent standby failure	Standby unavailable on demand
Bypass left active	Temporary impairment persists
Uncontrolled shutdown	Ceases without orderly preservation
Unsafe persistence	Remains active when hazardous
Leakage/escape	Fluid, energy, data or material escapes
Congestion/blockage	Flow is impeded
Loss of synchronisation	Coordinated state diverges
Incompatibility	Versions/materials/protocols conflict
Misallocation	Resource assigned wrongly
Unknown mode	Manner unresolved

Part VIII - Effects, propagation and consequence

Effect level	Question
Local	What happens at failed item/function?
Next-level	What happens to subsystem or dependent?
Cross-domain	Which other domain becomes impaired?
Capability	Which Service Outcome is degraded or lost?
Stakeholder	Who or what experiences consequence?

Operator	Use
CAUSES	Canonical causal, propagation, safeguard or evidence relationship.
CONTRIBUTES_TO	Canonical causal, propagation, safeguard or evidence relationship.
TRIGGERS	Canonical causal, propagation, safeguard or evidence relationship.
PROPAGATES_VIA	Canonical causal, propagation, safeguard or evidence relationship.
PROPAGATES_TO	Canonical causal, propagation, safeguard or evidence relationship.
AMPLIFIES	Canonical causal, propagation, safeguard or evidence relationship.
MASKS	Canonical causal, propagation, safeguard or evidence relationship.
BYPASSES	Canonical causal, propagation, safeguard or evidence relationship.
DEFEATS	Canonical causal, propagation, safeguard or evidence relationship.
CONTAINS	Canonical causal, propagation, safeguard or evidence relationship.
DELAYS	Canonical causal, propagation, safeguard or evidence relationship.
RECOVERS	Canonical causal, propagation, safeguard or evidence relationship.
REINTRODUCES	Canonical causal, propagation, safeguard or evidence relationship.
CORRELATES_WITH	Canonical causal, propagation, safeguard or evidence relationship.
CONFLICTS_WITH	Canonical causal, propagation, safeguard or evidence relationship.

Consequence family	Required expression
Life safety and health	Affected capability/stakeholder, magnitude, duration, concentration, assumptions and confidence.
Capability availability	Affected capability/stakeholder, magnitude, duration, concentration, assumptions and confidence.
Compute capacity	Affected capability/stakeholder, magnitude, duration, concentration, assumptions and confidence.
Performance	Affected capability/stakeholder, magnitude, duration, concentration, assumptions and confidence.

Consequence family	Required expression
Data/artifact integrity	Affected capability/stakeholder, magnitude, duration, concentration, assumptions and confidence.
Isolation and security	Affected capability/stakeholder, magnitude, duration, concentration, assumptions and confidence.
Physical damage	Affected capability/stakeholder, magnitude, duration, concentration, assumptions and confidence.
Environmental impact	Affected capability/stakeholder, magnitude, duration, concentration, assumptions and confidence.
Recovery complexity	Affected capability/stakeholder, magnitude, duration, concentration, assumptions and confidence.
Financial impact	Affected capability/stakeholder, magnitude, duration, concentration, assumptions and confidence.
Contractual/regulatory impact	Affected capability/stakeholder, magnitude, duration, concentration, assumptions and confidence.
Strategic/sovereignty impact	Affected capability/stakeholder, magnitude, duration, concentration, assumptions and confidence.
Evidence and attribution loss	Affected capability/stakeholder, magnitude, duration, concentration, assumptions and confidence.

Part IX - Safeguards and resilience

Function	Examples
Anticipate	Forecast, condition monitoring, capacity planning.
Prevent	Access control, design margin, protection, maintenance.
Detect	Sensors, telemetry, inspection and alarms.
Decide	Logic, operator, escalation and authority.
Respond	Transfer, isolation, throttling and dispatch.
Contain	Fault isolation, fire zone and segmentation.
Withstand	Reserve, hardening, redundancy and diversity.
Degrade gracefully	Priority, rate limiting and orderly shutdown.
Recover	Restart, rebuild, repair and data recovery.
Adapt	Redesign, learning and strengthened independence.

Barrier field	Required content
barrierId	Stable identifier
pathPosition	Cause, escalation, consequence or recovery side
objective	Path condition altered
dependencies	Power, sensing, control, people and supplier needs
applicableStates	States and transitions covered
capacity/duration	Demand it withstands
failureModes	How barrier can fail
validation	Test, inspection, event, analysis or simulation
confidence	Evidence-bounded state
residualExposure	Path remaining after demonstrated action

Safe validation hierarchy

Level	Method
V0	Document/configuration review
V1	Inspection/observation
V2	Historical event and telemetry
V3	Engineering analysis/simulation
V4	Tabletop/procedural exercise
V5	Component functional test
V6	Integrated systems test
V7	Live capability exercise where authorised and safe

Part X - Analysis techniques

Technique	AIDCF use
FMEA/FMECA	Selected according to purpose, lifecycle, complexity, evidence, state behaviour, safety and required output.
Fault Tree Analysis	Selected according to purpose, lifecycle, complexity, evidence, state behaviour, safety and required output.
Event Tree Analysis	Selected according to purpose, lifecycle, complexity, evidence, state behaviour, safety and required output.
Bow-tie Analysis	Selected according to purpose, lifecycle, complexity, evidence, state behaviour, safety and required output.
HAZOP-style study	Selected according to purpose, lifecycle, complexity, evidence, state behaviour, safety and required output.
What-if/SWIFT	Selected according to purpose, lifecycle, complexity, evidence, state behaviour, safety and required output.
Cause-consequence analysis	Selected according to purpose, lifecycle, complexity, evidence, state behaviour, safety and required output.
Dependency graph analysis	Selected according to purpose, lifecycle, complexity, evidence, state behaviour, safety and required output.
Common-cause analysis	Selected according to purpose, lifecycle, complexity, evidence, state behaviour, safety and required output.
State-transition analysis	Selected according to purpose, lifecycle, complexity, evidence, state behaviour, safety and required output.
Reliability block diagram	Selected according to purpose, lifecycle, complexity, evidence, state behaviour, safety and required output.
Root-cause analysis	Selected according to purpose, lifecycle, complexity, evidence, state behaviour, safety and required output.
Capability impact analysis	Selected according to purpose, lifecycle, complexity, evidence, state behaviour, safety and required output.
Scenario stress analysis	Selected according to purpose, lifecycle, complexity, evidence, state behaviour, safety and required output.
Validated digital-twin simulation	Selected according to purpose, lifecycle, complexity, evidence, state behaviour, safety and required output.

Default AIDCF hybrid: capability impact + dependency graph + selected FMEA/FMECA + top-event analysis for critical outcomes + barrier analysis + state-based scenario validation. No single technique is sufficient by default.

Part XI - Criticality and uncertainty

THF-REQ-022 Severity, likelihood, detectability and confidence shall remain separate.

THF-REQ-023 RPN or another arithmetic product shall not determine assurance by itself.

THF-REQ-024 Critical gates and catastrophic consequence shall not be averaged away.

THF-REQ-025 Quantitative probability shall be used only where data, model and assumptions are defensible.

Band	Capability interpretation
S0 Negligible	No material outcome effect.
S1 Minor	Local effect; outcome within requirement.
S2 Moderate	Recoverable degradation within approved tolerance.
S3 Major	Required outcome breached.
S4 Critical	Extended/concentrated loss or serious safety/strategic impact.
S5 Catastrophic	Fatality potential, mission loss or systemic irreversible consequence.

Uncertainty	Treatment
Model	Record alternate structures.
Parameter	Use range and sensitivity.
Evidence gap	Preserve UNKNOWN.
Temporal	State observation and validity.
Configuration	Separate intended, as-built, commissioned and observed.
Expert disagreement	Retain dissent and basis.
Rare event	Avoid false precision.
Emergent interaction	Use scenario, graph and stress analysis.

Part XII - Eight-domain failure catalogue

31. D1 Physical

Scenario family	Required analysis
Unauthorised entry	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Insider sabotage	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Surveillance blind spot	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Asset removal	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Security-zone collapse	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Emergency-access conflict	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.

32. D2 Facilities Engineering

Scenario family	Required analysis
Structural overload	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Fire-system mismatch	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Water ingress	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Common controller defect	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
As-built divergence	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Commissioning omission	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.

33. D3 Power & Cooling

Scenario family	Required analysis
Utility loss	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Generator no-start	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
UPS loss	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Protection miscoordination	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Rack overload	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Cooling capacity loss	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
CDU failure	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Thermal transient	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.

Scenario family	Required analysis
Liquid leak	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Shared control failure	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.

34. D4 Environmental & Resource

Scenario family	Required analysis
Ambient exceedance	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Water scarcity	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Poor coolant quality	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Air contamination	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Refrigerant loss	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Efficiency-resilience conflict	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Metering/claim error	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.

35. D5 Operations

Scenario family	Required analysis
Maintenance-induced outage	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Procedure failure	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Competence gap	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Change failure	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Alarm-response failure	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Spares/obsolescence	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Supplier support loss	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Handover failure	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.

36. D6 Compute

Scenario family	Required analysis
Accelerator/node failure	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Scale-up domain failure	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Firmware incompatibility	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.

Scenario family	Required analysis
Scheduler failure	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Telemetry blindness	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Thermal throttling	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Mass reboot	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Golden-image corruption	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.

37. D7 Storage

Scenario family	Required analysis
Metadata failure	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Capacity exhaustion	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Throughput collapse	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Checkpoint failure	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Protection common-mode	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Silent corruption	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Restore failure	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Data-transfer bottleneck	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.

38. D8 Network

Scenario family	Required analysis
Fabric partition	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Congestion collapse	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Scale-up link failure	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Management network loss	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Time-service failure	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
DNS/DHCP/IPAM failure	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Configuration blast radius	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.
Optical/cabling defect	Causes; functional mode; local and propagated effects; fault domains; safeguards; residual capability consequence; recovery; evidence and UNKNOWN.

Part XIII - Cross-domain scenario library

ID	Scenario	Minimum path analysis
X01	Utility loss + generator no-start + UPS autonomy exhaustion	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X02	Cooling loss at sustained AI load	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X03	Liquid leak with detection gap	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X04	Maintenance removes hidden shared dependency	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X05	BMS/EPMS compromise or malfunction	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X06	Fabric congestion during checkpoint demand	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X07	Storage metadata failure during training	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X08	Firmware rollout common-mode failure	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X09	Fire disables paired infrastructure routes	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X10	Extreme ambient plus water restriction	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X11	Operator error during alarm flood	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X12	Long-lead spare after rare failure	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X13	Time-source divergence	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X14	Colocation responsibility gap	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X15	Recovery reintroduces corrupt configuration	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X16	Physical intrusion disrupts interconnect	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X17	Expansion creates stranded compute	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X18	Loss of remote OEM support	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X19	Shared authentication service failure	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.
X20	Compound flood/fire/grid/telecom event	Initiating state → affected entities → dependencies → safeguards → residual capability → recovery → evidence.

Scenario prioritisation gates

- Life safety or hazardous energy
- Essential cooling loss
- Standby/resilient power loss
- Containment failure
- Large capability concentration
- Checkpoint or recovery-baseline corruption
- Common-mode defeat of redundancy

- Recovery objective at risk
- Weakly evidenced external dependency
- Long-lead or irreplaceable component
- Sovereignty impact
- Critical UNKNOWN

Part XIV - Assessment records and evidence

Step	Activity
1	Frame decision, capability, outcome, boundary and state
2	Confirm intended, as-built, commissioned and observed baseline
3	Map functions, assets, interfaces, dependencies and fault domains
4	Identify threats, hazards, causes and failure modes
5	Construct local, cross-domain, compound, maintenance and recovery scenarios
6	Analyse effects, propagation, timing and concentration
7	Map safeguards and their dependencies
8	Validate safely
9	Determine residual state, critical gates and UNKNOWN
10	Report finding, action, retest and reassessment triggers

Evidence class	Examples
Architecture	Single lines, P&IDs, topologies, control narratives.
As-built/configuration	Registers, settings, firmware, routes and dependencies.
Commissioning/testing	Functional, integrated, load, failover and recovery evidence.
Operations	Telemetry, alarms, trends, work orders and logs.
Incident	Timelines, cause analysis and near misses.
Supplier/contract	SLA, support, spares, lead time and responsibility.
Observation/interview	Walkthrough, witnessed action and competent testimony.
Engineering analysis	Studies, calculations and validated simulations.
Threat evidence	Site threat assessment, access, events and intelligence.

Finding taxonomy

Finding	Meaning
Uncontrolled path	Material consequence without effective barrier.
Fragile safeguard	Barrier has shared dependency, inadequate capacity or weak test.
Common-mode exposure	Alternatives share cause or dependency.
Containment gap	Effect crosses intended boundary.
Capacity cliff	Small change creates disproportionate loss.
Recovery fragility	Restoration depends on unproven state, people or resource.
Evidence gap	Material assertion inadequately supported.
State coverage gap	Degraded/maintenance/recovery state not covered.
Responsibility gap	Authority or evidence obligation unclear.
Critical UNKNOWN	Unresolved information caps assurance.

Part XV - Machine-readable model

Node	Purpose
ThreatSource	Canonical graph entity for AIDCF-004.
ThreatEvent	Canonical graph entity for AIDCF-004.
Hazard	Canonical graph entity for AIDCF-004.
InitiatingEvent	Canonical graph entity for AIDCF-004.
FailureCause	Canonical graph entity for AIDCF-004.
FailureMode	Canonical graph entity for AIDCF-004.
Effect	Canonical graph entity for AIDCF-004.
PropagationStep	Canonical graph entity for AIDCF-004.
Consequence	Canonical graph entity for AIDCF-004.
Safeguard	Canonical graph entity for AIDCF-004.
BarrierClaim	Canonical graph entity for AIDCF-004.
RecoveryAction	Canonical graph entity for AIDCF-004.
Scenario	Canonical graph entity for AIDCF-004.
EvidenceObject	Canonical graph entity for AIDCF-004.
Finding	Canonical graph entity for AIDCF-004.
AssuranceDecision	Canonical graph entity for AIDCF-004.

Relationship	Semantic use
TARGETS	Controlled directed relation; material assertions require provenance, state and confidence.
EXPOSES	Controlled directed relation; material assertions require provenance, state and confidence.
INITIATES	Controlled directed relation; material assertions require provenance, state and confidence.

Relationship	Semantic use
HAS_CAUSE	Controlled directed relation; material assertions require provenance, state and confidence.
FAILS_AS	Controlled directed relation; material assertions require provenance, state and confidence.
CAUSES	Controlled directed relation; material assertions require provenance, state and confidence.
CONTRIBUTES_TO	Controlled directed relation; material assertions require provenance, state and confidence.
TRIGGERS	Controlled directed relation; material assertions require provenance, state and confidence.
HAS_LOCAL_EFFECT	Controlled directed relation; material assertions require provenance, state and confidence.
PROPAGATES_VIA	Controlled directed relation; material assertions require provenance, state and confidence.
PROPAGATES_TO	Controlled directed relation; material assertions require provenance, state and confidence.
AMPLIFIED_BY	Controlled directed relation; material assertions require provenance, state and confidence.
AFFECTS	Controlled directed relation; material assertions require provenance, state and confidence.
PREVENTED_BY	Controlled directed relation; material assertions require provenance, state and confidence.
DETECTED_BY	Controlled directed relation; material assertions require provenance, state and confidence.
CONTAINED_BY	Controlled directed relation; material assertions require provenance, state and confidence.
RECOVERED_BY	Controlled directed relation; material assertions require provenance, state and confidence.
DEFEATS	Controlled directed relation; material assertions require provenance, state and confidence.
DEPENDS_ON	Controlled directed relation; material assertions require provenance, state and confidence.
EVIDENCED_BY	Controlled directed relation; material assertions require provenance, state and confidence.
CHALLENGED_BY	Controlled directed relation; material assertions require provenance, state and confidence.
RESULTS_IN	Controlled directed relation; material assertions require provenance, state and confidence.
INVALIDATES	Controlled directed relation; material assertions require provenance, state and confidence.

Publication package

- OWL ontology and Turtle serialisation
- JSON-LD context and examples
- JSON Schema for scenario exchange
- SHACL validation shapes
- CSV failure-mode and relationship registries
- Worked liquid-cooling and distributed-training examples
- Changelog, migration and extension guidance

Part XVI - Governance and acceptance

- Controlled vocabulary and change board
- Multidisciplinary domain review
- Semantic versioning and migration
- Normative core separated from informative examples
- Independent incident and test comparison
- Inter-assessor consistency testing
- Security classification of sensitive paths
- Vendor neutrality and conflict declarations

Prohibited practices

- Generic list treated as complete assessment

- Unsupported precise probability
- Average/RPN overriding critical gates
- Stopping at operator error
- Assuming duplicated paths are independent
- Untested safeguard treated as effective
- Ignoring recovery-only dependencies
- Unsafe intrusive validation
- Publishing sensitive paths without access control

Appendix A - Failure Mode Catalogue

ID	Group	Required function	Failure mode
FM-0001	Physical security	prevent entry	Complete loss
FM-0002	Physical security	prevent entry	Failure to start
FM-0003	Physical security	prevent entry	Failure to stop
FM-0004	Physical security	prevent entry	Failure to transfer
FM-0005	Physical security	prevent entry	Failure to isolate
FM-0006	Physical security	prevent entry	Failure to reset/recover
FM-0007	Physical security	prevent entry	Degraded output
FM-0008	Physical security	prevent entry	Excess output
FM-0009	Physical security	detect intrusion	Complete loss
FM-0010	Physical security	detect intrusion	Failure to start
FM-0011	Physical security	detect intrusion	Failure to stop
FM-0012	Physical security	detect intrusion	Failure to transfer
FM-0013	Physical security	detect intrusion	Failure to isolate
FM-0014	Physical security	detect intrusion	Failure to reset/recover
FM-0015	Physical security	detect intrusion	Degraded output
FM-0016	Physical security	detect intrusion	Excess output
FM-0017	Physical security	attribute access	Complete loss
FM-0018	Physical security	attribute access	Failure to start
FM-0019	Physical security	attribute access	Failure to stop
FM-0020	Physical security	attribute access	Failure to transfer
FM-0021	Physical security	attribute access	Failure to isolate
FM-0022	Physical security	attribute access	Failure to reset/recover
FM-0023	Physical security	attribute access	Degraded output
FM-0024	Physical security	attribute access	Excess output
FM-0025	Physical security	retain evidence	Complete loss
FM-0026	Physical security	retain evidence	Failure to start
FM-0027	Physical security	retain evidence	Failure to stop
FM-0028	Physical security	retain evidence	Failure to transfer
FM-0029	Physical security	retain evidence	Failure to isolate
FM-0030	Physical security	retain evidence	Failure to reset/recover
FM-0031	Physical security	retain evidence	Degraded output
FM-0032	Physical security	retain evidence	Excess output
FM-0033	Electrical	supply power	Complete loss
FM-0034	Electrical	supply power	Failure to start
FM-0035	Electrical	supply power	Failure to stop
FM-0036	Electrical	supply power	Failure to transfer
FM-0037	Electrical	supply power	Failure to isolate
FM-0038	Electrical	supply power	Failure to reset/recover
FM-0039	Electrical	supply power	Degraded output
FM-0040	Electrical	supply power	Excess output
FM-0041	Electrical	transfer source	Complete loss
FM-0042	Electrical	transfer source	Failure to start
FM-0043	Electrical	transfer source	Failure to stop
FM-0044	Electrical	transfer source	Failure to transfer
FM-0045	Electrical	transfer source	Failure to isolate
FM-0046	Electrical	transfer source	Failure to reset/recover
FM-0047	Electrical	transfer source	Degraded output
FM-0048	Electrical	transfer source	Excess output
FM-0049	Electrical	protect fault	Complete loss
FM-0050	Electrical	protect fault	Failure to start
FM-0051	Electrical	protect fault	Failure to stop
FM-0052	Electrical	protect fault	Failure to transfer
FM-0053	Electrical	protect fault	Failure to isolate
FM-0054	Electrical	protect fault	Failure to reset/recover
FM-0055	Electrical	protect fault	Degraded output
FM-0056	Electrical	protect fault	Excess output
FM-0057	Electrical	distribute power	Complete loss
FM-0058	Electrical	distribute power	Failure to start
FM-0059	Electrical	distribute power	Failure to stop
FM-0060	Electrical	distribute power	Failure to transfer
FM-0061	Electrical	distribute power	Failure to isolate
FM-0062	Electrical	distribute power	Failure to reset/recover
FM-0063	Electrical	distribute power	Degraded output
FM-0064	Electrical	distribute power	Excess output
FM-0065	Cooling	capture heat	Complete loss
FM-0066	Cooling	capture heat	Failure to start
FM-0067	Cooling	capture heat	Failure to stop
FM-0068	Cooling	capture heat	Failure to transfer
FM-0069	Cooling	capture heat	Failure to isolate
FM-0070	Cooling	capture heat	Failure to reset/recover
FM-0071	Cooling	capture heat	Degraded output
FM-0072	Cooling	capture heat	Excess output
FM-0073	Cooling	transport coolant	Complete loss
FM-0074	Cooling	transport coolant	Failure to start
FM-0075	Cooling	transport coolant	Failure to stop
FM-0076	Cooling	transport coolant	Failure to transfer
FM-0077	Cooling	transport coolant	Failure to isolate
FM-0078	Cooling	transport coolant	Failure to reset/recover
FM-0079	Cooling	transport coolant	Degraded output
FM-0080	Cooling	transport coolant	Excess output
FM-0081	Cooling	reject heat	Complete loss
FM-0082	Cooling	reject heat	Failure to start
FM-0083	Cooling	reject heat	Failure to stop
FM-0084	Cooling	reject heat	Failure to transfer
FM-0085	Cooling	reject heat	Failure to isolate
FM-0086	Cooling	reject heat	Failure to reset/recover
FM-0087	Cooling	reject heat	Degraded output
FM-0088	Cooling	reject heat	Excess output
FM-0089	Cooling	detect leak	Complete loss
FM-0090	Cooling	detect leak	Failure to start
FM-0091	Cooling	detect leak	Failure to stop

ID	Group	Required function	Failure mode
FM-0092	Cooling	detect leak	Failure to transfer
FM-0093	Cooling	detect leak	Failure to isolate
FM-0094	Cooling	detect leak	Failure to reset/recover
FM-0095	Cooling	detect leak	Degraded output
FM-0096	Cooling	detect leak	Excess output
FM-0097	Facilities OT	sense state	Complete loss
FM-0098	Facilities OT	sense state	Failure to start
FM-0099	Facilities OT	sense state	Failure to stop
FM-0100	Facilities OT	sense state	Failure to transfer
FM-0101	Facilities OT	sense state	Failure to isolate
FM-0102	Facilities OT	sense state	Failure to reset/recover
FM-0103	Facilities OT	sense state	Degraded output
FM-0104	Facilities OT	sense state	Excess output
FM-0105	Facilities OT	execute logic	Complete loss
FM-0106	Facilities OT	execute logic	Failure to start
FM-0107	Facilities OT	execute logic	Failure to stop
FM-0108	Facilities OT	execute logic	Failure to transfer
FM-0109	Facilities OT	execute logic	Failure to isolate
FM-0110	Facilities OT	execute logic	Failure to reset/recover
FM-0111	Facilities OT	execute logic	Degraded output
FM-0112	Facilities OT	execute logic	Excess output
FM-0113	Facilities OT	command actuator	Complete loss
FM-0114	Facilities OT	command actuator	Failure to start
FM-0115	Facilities OT	command actuator	Failure to stop
FM-0116	Facilities OT	command actuator	Failure to transfer
FM-0117	Facilities OT	command actuator	Failure to isolate
FM-0118	Facilities OT	command actuator	Failure to reset/recover
FM-0119	Facilities OT	command actuator	Degraded output
FM-0120	Facilities OT	command actuator	Excess output
FM-0121	Facilities OT	raise alarm	Complete loss
FM-0122	Facilities OT	raise alarm	Failure to start
FM-0123	Facilities OT	raise alarm	Failure to stop
FM-0124	Facilities OT	raise alarm	Failure to transfer
FM-0125	Facilities OT	raise alarm	Failure to isolate
FM-0126	Facilities OT	raise alarm	Failure to reset/recover
FM-0127	Facilities OT	raise alarm	Degraded output
FM-0128	Facilities OT	raise alarm	Excess output
FM-0129	Operations	authorise work	Complete loss
FM-0130	Operations	authorise work	Failure to start
FM-0131	Operations	authorise work	Failure to stop
FM-0132	Operations	authorise work	Failure to transfer
FM-0133	Operations	authorise work	Failure to isolate
FM-0134	Operations	authorise work	Failure to reset/recover
FM-0135	Operations	authorise work	Degraded output
FM-0136	Operations	authorise work	Excess output
FM-0137	Operations	isolate energy	Complete loss
FM-0138	Operations	isolate energy	Failure to start
FM-0139	Operations	isolate energy	Failure to stop
FM-0140	Operations	isolate energy	Failure to transfer
FM-0141	Operations	isolate energy	Failure to isolate
FM-0142	Operations	isolate energy	Failure to reset/recover
FM-0143	Operations	isolate energy	Degraded output
FM-0144	Operations	isolate energy	Excess output
FM-0145	Operations	respond	Complete loss
FM-0146	Operations	respond	Failure to start
FM-0147	Operations	respond	Failure to stop
FM-0148	Operations	respond	Failure to transfer
FM-0149	Operations	respond	Failure to isolate
FM-0150	Operations	respond	Failure to reset/recover
FM-0151	Operations	respond	Degraded output
FM-0152	Operations	respond	Excess output
FM-0153	Operations	recover	Complete loss
FM-0154	Operations	recover	Failure to start
FM-0155	Operations	recover	Failure to stop
FM-0156	Operations	recover	Failure to transfer
FM-0157	Operations	recover	Failure to isolate
FM-0158	Operations	recover	Failure to reset/recover
FM-0159	Operations	recover	Degraded output
FM-0160	Operations	recover	Excess output
FM-0161	Compute	boot node	Complete loss
FM-0162	Compute	boot node	Failure to start
FM-0163	Compute	boot node	Failure to stop
FM-0164	Compute	boot node	Failure to transfer
FM-0165	Compute	boot node	Failure to isolate
FM-0166	Compute	boot node	Failure to reset/recover
FM-0167	Compute	boot node	Degraded output
FM-0168	Compute	boot node	Excess output
FM-0169	Compute	execute workload	Complete loss
FM-0170	Compute	execute workload	Failure to start
FM-0171	Compute	execute workload	Failure to stop
FM-0172	Compute	execute workload	Failure to transfer
FM-0173	Compute	execute workload	Failure to isolate
FM-0174	Compute	execute workload	Failure to reset/recover
FM-0175	Compute	execute workload	Degraded output
FM-0176	Compute	execute workload	Excess output
FM-0177	Compute	coordinate accelerators	Complete loss
FM-0178	Compute	coordinate accelerators	Failure to start
FM-0179	Compute	coordinate accelerators	Failure to stop
FM-0180	Compute	coordinate accelerators	Failure to transfer
FM-0181	Compute	coordinate accelerators	Failure to isolate
FM-0182	Compute	coordinate accelerators	Failure to reset/recover
FM-0183	Compute	coordinate accelerators	Degraded output
FM-0184	Compute	coordinate accelerators	Excess output
FM-0185	Compute	schedule jobs	Complete loss
FM-0186	Compute	schedule jobs	Failure to start

ID	Group	Required function	Failure mode
FM-0187	Compute	schedule jobs	Failure to stop
FM-0188	Compute	schedule jobs	Failure to transfer
FM-0189	Compute	schedule jobs	Failure to isolate
FM-0190	Compute	schedule jobs	Failure to reset/recover
FM-0191	Compute	schedule jobs	Degraded output
FM-0192	Compute	schedule jobs	Excess output
FM-0193	Storage	serve data	Complete loss
FM-0194	Storage	serve data	Failure to start
FM-0195	Storage	serve data	Failure to stop
FM-0196	Storage	serve data	Failure to transfer
FM-0197	Storage	serve data	Failure to isolate
FM-0198	Storage	serve data	Failure to reset/recover
FM-0199	Storage	serve data	Degraded output
FM-0200	Storage	serve data	Excess output
FM-0201	Storage	maintain metadata	Complete loss
FM-0202	Storage	maintain metadata	Failure to start
FM-0203	Storage	maintain metadata	Failure to stop
FM-0204	Storage	maintain metadata	Failure to transfer
FM-0205	Storage	maintain metadata	Failure to isolate
FM-0206	Storage	maintain metadata	Failure to reset/recover
FM-0207	Storage	maintain metadata	Degraded output
FM-0208	Storage	maintain metadata	Excess output
FM-0209	Storage	persist checkpoint	Complete loss
FM-0210	Storage	persist checkpoint	Failure to start
FM-0211	Storage	persist checkpoint	Failure to stop
FM-0212	Storage	persist checkpoint	Failure to transfer
FM-0213	Storage	persist checkpoint	Failure to isolate
FM-0214	Storage	persist checkpoint	Failure to reset/recover
FM-0215	Storage	persist checkpoint	Degraded output
FM-0216	Storage	persist checkpoint	Excess output
FM-0217	Storage	restore data	Complete loss
FM-0218	Storage	restore data	Failure to start
FM-0219	Storage	restore data	Failure to stop
FM-0220	Storage	restore data	Failure to transfer
FM-0221	Storage	restore data	Failure to isolate
FM-0222	Storage	restore data	Failure to reset/recover
FM-0223	Storage	restore data	Degraded output
FM-0224	Storage	restore data	Excess output
FM-0225	Network	forward traffic	Complete loss
FM-0226	Network	forward traffic	Failure to start
FM-0227	Network	forward traffic	Failure to stop
FM-0228	Network	forward traffic	Failure to transfer
FM-0229	Network	forward traffic	Failure to isolate
FM-0230	Network	forward traffic	Failure to reset/recover
FM-0231	Network	forward traffic	Degraded output
FM-0232	Network	forward traffic	Excess output
FM-0233	Network	control congestion	Complete loss
FM-0234	Network	control congestion	Failure to start
FM-0235	Network	control congestion	Failure to stop
FM-0236	Network	control congestion	Failure to transfer
FM-0237	Network	control congestion	Failure to isolate
FM-0238	Network	control congestion	Failure to reset/recover
FM-0239	Network	control congestion	Degraded output
FM-0240	Network	control congestion	Excess output
FM-0241	Network	provide timing	Complete loss
FM-0242	Network	provide timing	Failure to start
FM-0243	Network	provide timing	Failure to stop
FM-0244	Network	provide timing	Failure to transfer
FM-0245	Network	provide timing	Failure to isolate
FM-0246	Network	provide timing	Failure to reset/recover
FM-0247	Network	provide timing	Degraded output
FM-0248	Network	provide timing	Excess output
FM-0249	Network	enable management	Complete loss
FM-0250	Network	enable management	Failure to start
FM-0251	Network	enable management	Failure to stop
FM-0252	Network	enable management	Failure to transfer
FM-0253	Network	enable management	Failure to isolate
FM-0254	Network	enable management	Failure to reset/recover
FM-0255	Network	enable management	Degraded output
FM-0256	Network	enable management	Excess output

Appendix B - Working templates

[Context/state], [threat/hazard/event] causes [mode] in [entity/function]. Effect propagates through [dependencies/fault domains] and affects [Service Outcome]. [Safeguards] are expected to act. Evidence [supports/challenges/is insufficient]. Residual state is [state], with [confidence/UNKNOWN].

Step	Entity	Pre-state	Event/mode	Dependency	Effect	Safeguard	Post-state	Evidence
1								
2								
3								
4								

Step	Entity	Pre-state	Event/mode	Dependency	Effect	Safeguard	Post-state	Evidence
5								
6								
7								
8								

Function	Mode	Cause	Local effect	End effect	Safeguards	Evidence	Criticality	Action

Appendix C - References

Reference	Use
IEC 60812:2018	FMEA/FMECA planning, execution, documentation and maintenance context.
IEC 31010:2019	Selection and application of risk-assessment techniques.
NIST SP 800-30 Rev. 1	Threat-oriented risk-assessment comparison.
AIDCF Manifesto v1.0	Doctrine, evidence, UNKNOWN and critical-gate principles.
AIDCF Core Conceptual Model v1.0	Canonical failure-propagation concepts.
AIDCF-003 v1.0	Entities, dependencies, states, capacity and fault domains.
ASHRAE TC 9.9 guidance	Data-centre thermal and equipment-environment context.

Appendix D - Review and change record

Review role	Status	Focus
Methodology author	Draft completed	Discipline validity, boundaries, evidence and publication readiness.
Reliability/FMEA	Pending	Discipline validity, boundaries, evidence and publication readiness.
Data-centre architecture	Pending	Discipline validity, boundaries, evidence and publication readiness.
Electrical	Pending	Discipline validity, boundaries, evidence and publication readiness.
Mechanical/liquid cooling	Pending	Discipline validity, boundaries, evidence and publication readiness.
Fire/life safety	Pending	Discipline validity, boundaries, evidence and publication readiness.
Physical security	Pending	Discipline validity, boundaries, evidence and publication readiness.
AI compute/HPC	Pending	Discipline validity, boundaries, evidence and publication readiness.
Storage	Pending	Discipline validity, boundaries, evidence and publication readiness.
Network/fabric	Pending	Discipline validity, boundaries, evidence and publication readiness.
OT security	Pending	Discipline validity, boundaries, evidence and publication readiness.
Operations/recovery	Pending	Discipline validity, boundaries, evidence and publication readiness.
Ontology/graph	Pending	Discipline validity, boundaries, evidence and publication readiness.
Legal/IP	Pending	Discipline validity, boundaries, evidence and publication readiness.
Public release	Not approved	Mandatory gates closed.

Version	Status	Summary
0.1	Architecture draft	Ontology and doctrine.

Version	Status	Summary
0.5	Integrated draft	Domain catalogues, scenarios and graph semantics.
1.0	Controlled-review draft	Comprehensive baseline for expert review.

Final doctrine

Do not ask only whether a control exists or a component is redundant. Ask what the capability requires, what can disturb the required conditions, how functions can fail, where effects propagate, which barriers remain available in the applicable state, what evidence demonstrates their behaviour, what consequence remains, how recovery is achieved and what is still UNKNOWN.